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# Behavioral Analysis of Information Salience in Large Language Models

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## Abstract

Large Language Models (LLMs) excel at text summarization, a task that requires models to select content based on its importance. However, the exact notion of salience that LLMs have internalized remains unclear. To bridge this gap, we introduce an explainable framework to systematically derive and investigate information salience in LLMs through their summarization behavior. Using length-controlled summarization as a behavioral probe into the content selection process, and tracing the answerability of Questions Under Discussion throughout, we derive a proxy for how models prioritize information. Our experiments on 13 models across four datasets reveal that LLMs have a nuanced, hierarchical notion of salience, generally consistent across model families and sizes. While models show highly consistent behavior and hence salience patterns, this notion of salience cannot be accessed through introspection, and only weakly correlates with human perceptions of information salience.[1]

## 1 Introduction

Large Language Models (LLMs) significantly advanced text synthesis tasks, including text summarization, which they perform well even under zero-shot conditions (Goyal et al., 2023; Zhang et al., 2024). The nature of the summarization task requires models to do content selection: picking the most salient pieces of information for inclusion in a summary (Mani and Maybury, 1999). However, it remains unclear what underlying notion of salience the models have internalized.

Prior work investigated information salience from several angles. Theories of discourse structure have been used to induce content salience (Marcu, 1999; Louis et al., 2010), and a large body of summarization research uses word distribution or centrality as the main signal for content selection (Nenkova and McKeown, 2012; Nazari and Mahdavi, 2019). Peyrard (2019) laid out a theoretical perspective for content salience in summarization, though the exact notion remains largely latent and aloof; rather, pre-LLM summarization work uses human summaries as supervision signals to learn what to include (Gehrmann et al., 2018; Chen and Bansal, 2018; Liu and Lapata, 2019, *inter alia*). Yet, none of these accounts explains why LLM zero-shot summarization works so well on the one hand, while missing key elements on the other (Kim et al., 2024; Trienes et al., 2024; Huang et al., 2024).

To begin to make sense of this behavior, we need to understand how models internalize salience: whether it is a *consistent* notion within and across models, *how* they prioritize information, and whether LLMs' notion of salience *aligns* with prior theories or human intuitions.

In this paper, we present a novel explainable framework to systematically derive and investigate LLMs' grasp of information salience through their summarization behavior. Our method combines **two key ideas**. First, we can use length-constrained summarization (Fan et al., 2018; He et al., 2022) as a behavioral probe into the content selection process of LLMs. Intuitively, when there is a limited length budget for a summary, we posit that the least important information is dropped first.

Second, we can describe what is salient as the answerability of domain-relevant Questions Under Discussion (QUDs; Van Kuppevelt, 1995; Benz and Jasinskaja, 2017; Wu et al., 2023a). QUDs can be thought of as representations of a coherent unit of information in the form of information-seeking questions, e.g., *Who are the participants of this study?* We use such questions — and hence their answers extracted from documents, according to alternative semantics (Hamblin, 1973; Karttunen, 1977; Groenendijk and Stokhof, 1984) — as the

[1]We release code, model outputs and human annotations at https://github.com/jantrienes/llm-salience.

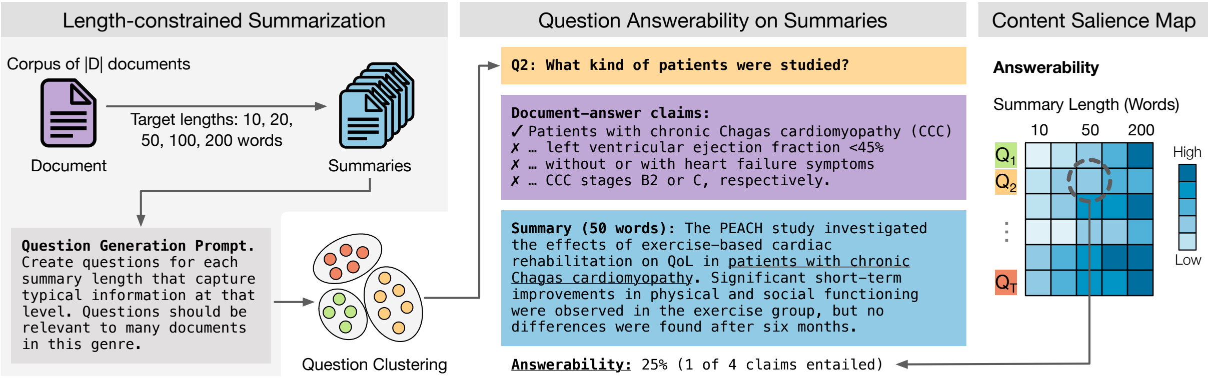


Figure 1: Framework overview, conceptualizing content salience as question answerability. **Left:** Given a corpus, we derive questions that are typically answered in summaries. Length-controlled summarization acts as a probe into the content-selection process of LLMs. Question paraphrases are clustered by semantic intent. **Middle:** Answerability is calculated as the fraction of document-answer claims entailed by the summary. **Right:** The content salience map tracks answerability at each summary length. More salient questions remain answerable even in shorter summaries.

primary unit of analysis, making our framework interpretable and customizable.

Taken together, by gradually decreasing the length budget available for a summary and by systematically tracing question answerability throughout, we can derive a *proxy* for how models prioritize information. See Figure 1 for an overview.

Using this framework (§ 2), we empirically study LLMs' content selection *behavior*, and its alignment with *perceived* notions of salience. Through experiments on 13 models and four datasets (§ 3), we aim to answer the following research questions:

**RQ1** What notion of salience have LLMs learned in different domains?

**RQ2** Do LLMs of different families/sizes have a similar notion of salience?

**RQ3** When models introspect, does their perceived notion of salience align with their summarization behavior?

**RQ4** To what extent does model salience align with human perceived salience?

We find that LLMs have a nuanced notion of salience prioritizing information hierarchically across summary lengths (§ 4.1). Also the notion of salience is generally compatible between models even of different families and sizes, though more recent/bigger LLMs correlate more strongly with GPT-4o (§ 4.2). Furthermore, models show highly consistent behavior and hence notions of salience, but it cannot be elicited through introspection (i.e., directly prompting for salience of topics; § 5.2). Lastly, we find that model behavior only weakly aligns with human perceptions of salience (§ 5.2).

## 2 Method: Analyzing Content Salience

To analyze content salience we need a way to both *observe* what content models consider important (§ 2.1), and to *describe* it in an interpretable manner (§ 2.2). Figure 1 illustrates the framework.

### 2.1 Length-constrained Summarization as a Content Salience Probe

To elicit content-selection decisions from models, we use length-constrained summarization as a probe. Our key intuition is that, under a limited length budget, well-behaving models will drop the least important information first, while preserving the most salient content.

**Summary Generation.** Given a corpus $D$, and a set of target lengths $L$ specified in words, we generate summaries $S = \{s_{d,l} \mid d \in D, l \in L\}$ for all documents and length targets. We consider $L = \{10, 20, 50, 100, 200\}$ to capture a range of typical summary lengths.

**Tracing Content-selection Decisions.** To understand how summary content changes with varying length budgets, we introduce *Content Salience Maps (CSMs)* as a structured representation to systematically track the inclusion and exclusion of topics. Formally, let $T$ be a set of topics of interest, and let $f : T \times S \to [0, 1]$ be a function that measures to what extent topic $t$ is present in summary $s$. For a document $d$, CSM($d$) is a $|T| \times |L|$ matrix, where each entry is defined as:

$$\text{CSM}(d)_{t,l} = f(t, s_{d,l}). \tag{1}$$

We define the corpus-level CSM$(D)$ as the average of document-level measurements:

$$\text{CSM}(D)_{t,l} = \frac{1}{|D^t|} \sum_{d \in D^t} f(t, s_{d,l}), \tag{2}$$

where $D^t$ is the set of documents that contain topic $t$. We also define *topic prevalence* as $|D^t|/|D|$, representing the fraction of documents in the corpus that contain topic $t$.

Below, we describe a concrete instantiation of this framework, where the set of topics $T$ is represented as QUDs, and the inclusion measure $f$ is defined as question answerability. However, we note that the framework is highly customizable in terms of the definitions of $T$ and $f$.

## 2.2 Question-based Content Analysis

We represent the topics $t \in T$ as Questions Under Discussion (QUD), a linguistic representation for topics in discourse (Van Kuppevelt, 1995). In our setup, each QUD represents a possible *answer space* across different documents in the same genre. This aligns with alternative semantics, where questions are viewed as the set of possible answers (Hamblin, 1973; Karttunen, 1977; Groenendijk and Stokhof, 1984). In addition to the interpretability provided by natural language questions, we can also quantify content salience through *question answerability*: questions which remain answerable even with shorter summaries are more salient than questions which can only be answered with longer summaries. Below (also Algorithm 1), we describe a four-step pipeline to implement this approach.

**Step 1: Question Generation.** We design a question-generation prompt inspired by (Laban et al., 2022). Given summaries of varying lengths from a random sample of documents, we prompt an LLM to generate $n$ questions which each summary answers in a unique way. The prompt specifies two requirements: (1) the questions should be answerable by most documents in the given genre, and (2) they should highlight meaningful differences between summaries of different lengths (full prompt in Appendix G). For example, in movie reviews, most summaries will answer questions such as *"What is the main plot of the movie?"*, but naturally the answers will be different for each review. We repeat this process for all documents and associated summaries in the corpus.[2]

[2] We ran question generation over GPT-4o, Llama 3.1 (8B), and Mistral summaries. As the resulting questions were highly similar, we did not include additional models.

**Algorithm 1** Content Salience Map (CSM) Derivation

**Input:** Corpus: $D = \{d_1, d_2, ..., d_{|D|}\}$
Lengths: $L = \{10, 20, 50, 100, 200\}$ (words)
Models: $M_{\text{Sum}}, M_{\text{QG}}, M_{\text{Emb}}, M_{\text{QA}}, M_{\text{ClaimSplit}}, M_{\text{NLI}}$
**Output:** Corpus-level CSM$_D$

**for** $(d, l) \in D \times L$ **do** ▷ *Step 0: Summarization*
  $S[d, l] \leftarrow M_{\text{Sum}}(d, l)$
**for** $d \in D$ **do** ▷ *Step 1: Question Generation*
  $Q \leftarrow Q \cup M_{\text{QG}}(d, S[d, :])$
$T \leftarrow \text{Cluster}(M_{\text{Emb}}(Q))$ ▷ *Step 2: Question Clustering*
$T \leftarrow \text{ManualReview}(T)$
$T \leftarrow \text{SelectClusterRepresentatives}(T)$
**for** $(d, t) \in D \times T$ **do** ▷ *Step 3: QA and Claim Split*
  $\text{ans}_{\text{ref}} \leftarrow M_{\text{QA}}(d, t)$
  **if** $\text{ans}_{\text{ref}} \neq \varnothing$ **then** $A[d, t] \leftarrow M_{\text{ClaimSplit}}(\text{ans}_{\text{ref}})$
  **else** $A[d, t] \leftarrow \varnothing$
**for** $(t, l) \in T \times L$ **do** ▷ *Step 4: Answerability*
  **for** $d \in D$ **do**
    $s, A_t \leftarrow S[d, l], A[d, t]$ ▷ *(summary, claims)*
    $\text{CSM}_d[t, l] \leftarrow \text{avg}([M_{\text{NLI}}(a, s) \mid a \in A_t])$ ▷ *Eq. 3*
  $D_t \leftarrow \{d \in D \wedge A[d, t] \neq \varnothing\}$
  $\text{CSM}_D[t, l] \leftarrow \text{avg}([\text{CSM}_d[t, l] \mid d \in D_t])$ ▷ *Eq. 2*
**return** CSM$_D$

**Step 2: Clustering.** We then cluster questions with the same semantic intent. For instance, *"Is the soundtrack effective?"* and *"How does the music contribute to the film's atmosphere?"* are considered equivalent, as they ask for the same information. We select the question closest to the mean embedding of each cluster as its representative. These questions form the topics $T$.

**Step 3: Question-Answering and Claim Decomposition.** For each `(original document, question)` pair, we first obtain a *reference answer* using a QA-model. We then decompose each answer into a set of atomic claims $A_t$ (see Figure 1 for an example). These claims support the answerability calculation (described next), and a fine-grained analysis of summary similarity and consistency through claim entailment patterns (§ 4.2).

**Step 4: Answerability Estimation.** We measure how well a summary answers a question by the fraction of reference answer claims it entails. This naturally accounts for questions that are only partly answerable with a given summary. Formally, let $A_t$ be the set of answer claims for a given question. The answerability score is then calculated as:

$$f(t, s) = \frac{1}{|A_t|} \sum_{a \in A_t} e(a, s), \tag{3}$$

where $e : A \times S \rightarrow \{0, 1\}$ is a natural language inference (NLI) model that determines if claim $a$ is entailed (1) or not entailed (0) by summary $s$. This